

Is the 360° street view redundant?

Phase-2 attention pruning — Qwen3.6-27B, $k=1$, $n=1663$ solved questions

The claim under test

Each urban VQA question shows **4 street-view cameras** (fwd, bwd, cross_left, cross_right).

Claim

For 8 urban-attribute MCQ tasks, the full 360° view is **redundant**: the **1 image the model attends to most** keeps the answer correct — and that image is the **forward camera**.

Setup. Solve on 4 images → keep only $k=1$ image per arm → re-ask greedily. **Survival %** = of questions solved on 4 images, fraction still correct on the kept image (a per-question *ceiling*, never a raw accuracy).

Arms: **top** (most-attended) | **fwd** (always forward) | random | **bottom** (least-attended) | zero (no image = leakage probe).

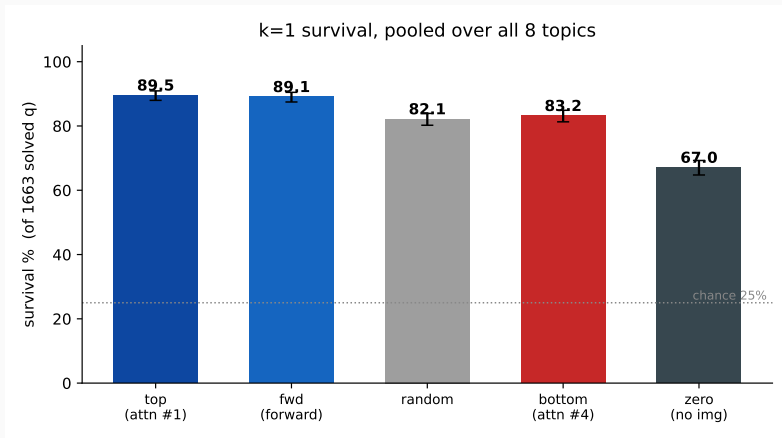
Verdict in one slide

Claim	Verdict	Key number
A. 360° redundant; <i>one</i> image often enough	supported	top 89.5% [88.0, 90.9]; even random 82.1%
B. Attention <i>ranking</i> selects the right image	partial	top—random +7.4 [+5.0, +9.8], but top $\approx$ fwd
C. The image that matters is the <i>forward</i> camera	supported	fwd 89.1% [87.5, 90.5]; attn. #1 = fwd 94.8%

One line

The 360° view **is** redundant and a **single forward image** usually suffices. Attention beats random selection — but only because it **almost always points at the forward camera**. No evidence (at this power) that the *ranking* adds anything beyond “use forward.”

Pooled survival — one image is usually enough



`top=fwd` $\gg$ `random` $\approx$ `bottom` $\gg$ `zero`. Even the *least*-attended image survives 83.2% — “even the worst single view is usually enough.”

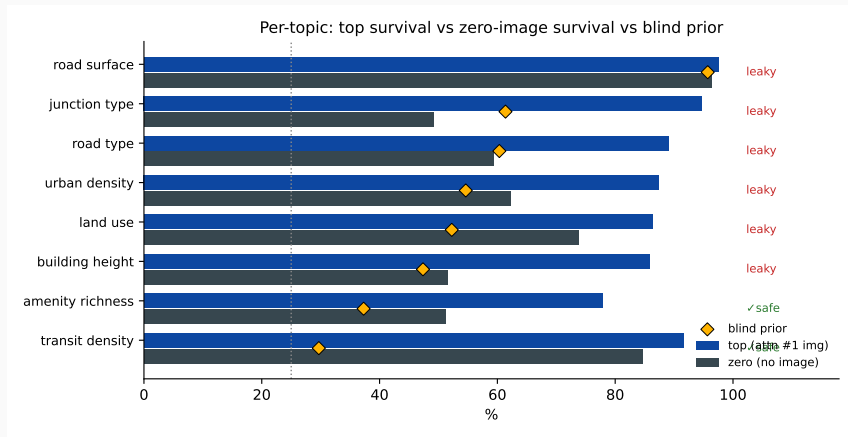
The leakage problem dominates everything

6 of 8 topics are answerable from **text + options alone**. The **zero** arm (no image) is the empirical detector:

topic	blind	zero-surv	topic	blind	zero-surv
road_surface	0.957	96.4%	land_use	0.523	73.8%
junction_type	0.614	49.1%	amenity_richness	0.373	51.3%
road_type	0.603	59.4%	transit_density	0.297	84.6%(!)
urban_density	0.546	62.3%	building_height	0.474	51.5%

- Pooled **zero**-image survival is **67%** — most “survival” is *not* image use.
- Only **amenity_richness** and **transit_density** (blind < 0.45) carry the headline. The other six are reported but **excluded**.
- Even a “safe” topic (transit) survives 84.6% with **zero** images.

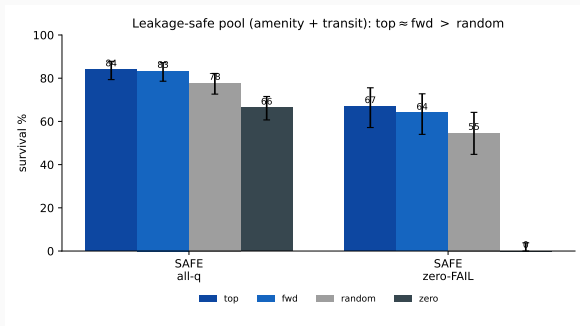
Per-topic — top vs zero vs blind prior



Sorted by blind prior. road_surface: top survival rides almost entirely on the 0.957 prior (zero already 96.4%) — **no** image signal.

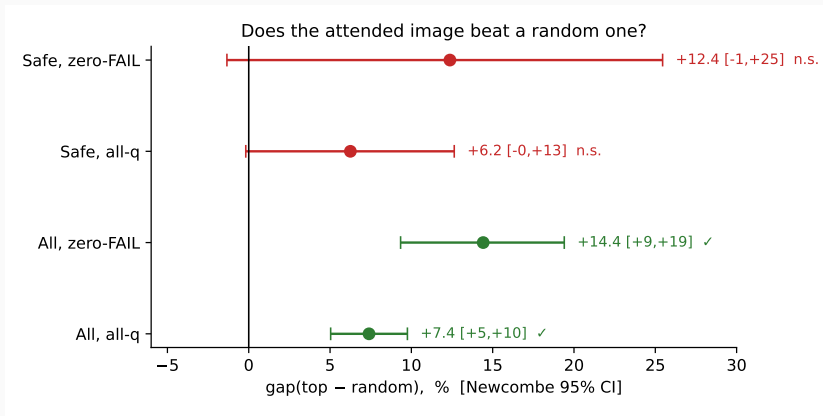
Honest headline: condition on questions that **NEED** an image

Cleanest test — restrict to the **zero-FAIL** subset (model wrong with no image), leakage-safe pool:



- top $\approx$ fwd clearly above random/zero — redundancy + “forward matters” **hold** on the clean pool.
- **But** gap(top–random) = +6.2 [-0.2, +12.6] (all-q), +12.4 [-1.3, +25.4] (zero-FAIL): **CI includes 0** — **not significant** (n=288/97).

Does the attended image beat a random one?



Gap is **real & significant on the full pool** (and *grows* on zero-FAIL: leakage doesn't explain it). On the clean 2-topic safe pool it is **only suggestive** — small n .

top vs fwd: does attention earn its keep?

Attention's #1 image is **the forward camera in** 94.8% of questions. It disagrees in only **86 / 1663**.

scope	top	fwd	gap(top-fwd)
all pool ($n=1663$)	89.5%	89.1%	+0.5 [-1.6, +2.6] n.s.
attn. $\neq$ fwd ($n=86$)	82.6%	73.3%	+9.3 [-3.2, +21.4] n.s.

Honest reading

The “attention selects the right image” claim **collapses into** “keep the forward image.” When attention *disagrees* with forward it *may* win (+9.3 pts) — but $n=86$ is too small to prove it.

That 86-question subset is where a larger run earns the stronger claim.

Adversarial checks (tried to break it)

- **Leakage cancellation?** No — gap *grows* on image-dependent subset (+7.4 $\rightarrow$ +14.4). Real signal.
- **bottom $\geq$ random?** Tie: +1.0 [-1.6, +3.6]. Verified **not** a bug — bottom kept the rank-4 image in **1663/1663**. Supports redundancy.
- **Position confound.** fwd kept partly for its fixed grid slot; cannot separate content from position. **Stated, not resolved** (fwd success ranges 51–98% across topics $\Rightarrow$ tracks content somewhat).
- **CI discipline.** Wilson/Newcombe on every rate & gap; CI-includes-0 marked n.s.
- **k -monotonicity.** **Untestable** — only $k=1$ complete.

Limitations (be skeptical)

1. **Leakage dominates** — 6/8 topics text-solvable; pooled zero-survival 67%. Headline rests on 2 topics, where the gap is *not* significant.
2. **$k=1$ only** — no $k=2/3$ curve; the “1–3 images” claim is half-tested. (Multi-seed/ k run is 2.1% done $\rightarrow$ unusable.)
3. **Single random seed (3407)** — no per-seed spread.
4. **Position confound** on the forward camera.
5. **Solver = 27B $\neq$ deployed model** — survival is a ceiling for this solver.
6. **Selection bias** — pool = questions solved on 4 images (conditional).
7. **Single-token attention** — `choice` $\equiv$ `avg`; ranking is one across-layer mean, a coarse signal.

- ✓ 360° is **redundant**; **one** street-view image usually suffices.
- ✓ That image is the **forward camera** (`along_fwd`).
- ~ Attention **beats random**, but mostly *because* it picks forward — “attention selects the right image” is **not** yet earned.

Repro: `python attention_prune/analyze_prune_k1.py` | analysis sha `b083f41` | `n=1663`, seed 3407.